

Does Information-Gain Feature Selection Preserve Classifier Accuracy?

A Comparative Study on Mushroom Edibility Classification

Course Mini-Project Report

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The dataset will often have a lot more features than what the model requires. It is thus important to determine which features can be considered for use. This study tries to evaluate whether the amount of information gained by each feature from an entropy perspective is a good indicator of whether that feature should be used in building a model. In this project, the UCI Mushroom dataset (8124 mushrooms, 22 categorical features, edible/poisonous labeled) is used. Information gain is then used to rank the features. Logistic Regression classifier and Decision Tree classifier is then trained using both the whole set of 22 features as well as the top 5 features. The loss of accuracy when using less features is at most 0.1 percentage point. Moreover, the ranking of the information gain corresponds very well to the feature chosen by the decision tree as the first split.

1. Introduction and Background

Datasets are common with features, but not all are equally useful to a model. Choosing which inputs to use in a model is often referred to as a feature selection problem, because simpler models tend to be faster to train, easier to interpret, and more likely to generalize to new data (as opposed to overfitting their training set). Information gain (equivalently, mutual information between a feature and the target) is one way to measure a feature's usefulness, and is the criterion decision trees use internally to choose what split to make. This project tests this hypothesis on a mushroom classification dataset (edible vs. poisonous), a fully categorical dataset well suited to entropy and information based analysis to see if applying information theory to understand the data has any bearing on the results we get from other algorithms.

1.1 Entropy and Information Gain

Entropy $H = -\sum_i p_i \log_2 p_i$ is a measure of uncertainty in a target variable. For a binary target, $H = -p_1 \log_2 p_1 - p_0 \log_2 p_0$ is maximized at 1 bit in the case of a balanced (50/50) outcome and 0 if the outcome is always known. Information gain for a feature is the reduction in entropy achieved by splitting on that feature's values, this is the quantity decision trees internally use to decide which feature to split on first, and is mathematically equivalent to the mutual information between that feature and the target in the discrete case.

1.2 Logistic Regression

Logistic regression fits a linear decision boundary in log-odds space, where each feature is assigned a weight; the sign and magnitude of each weight indicate how much and in what direction a feature pushes the prediction. The raw linear value (the log-odds) is then transformed into a probability between 0 and 1 using the sigmoid function, $\text{sigmoid}(X) = 1 / (1 + e^{-X})$.

1.3 Decision Trees

Figure 1. The sigmoid function, which logistic regression uses to convert a raw log-odds score into a probability.

Decision trees recursively split the data on features, up to a specified maximum depth (to avoid overfitting which is memorizing noise in the training data rather than learning patterns that generalize to new data).

2. Problem Statement / Research Question

Does choosing features based on information gain result in a model with accuracy comparable to a model trained on all features, for the mushroom edibility classification dataset?

3. Objectives

- Rank all the features in the dataset by information gain with respect to the edibility label.
- Train and evaluate two classifiers — logistic regression and a decision tree — on both the full feature set and a reduced top-k feature set.
- See whether the reduced feature models perform comparably (in terms of accuracy, precision, and recall) to the full feature models.
- See if the information gain ranking matches up with the splits the decision tree actually learns.

4. Methodology

The dataset (8,124 rows, 22 categorical features plus the edibility label) was loaded with pandas. The only missing values occur in stalk root and are marked '?' (2,480 of 8,124 rows, 30.5%); these were kept as their own category to avoid waste, as discarding that many rows would be wasteful and the missingness is tied to the species rather than occurring at random.

The data was split 75/25 into training and test sets (stratified by class, random_state=42 for reproducibility), and every subsequent computation (entropy, information gain, and model fitting) was done only on the training split so the held-out test set didn't accidentally leak any information into the feature selection process.

Information gain was computed for each of the 22 original categorical features against the training set using the entropy formula above, then the features were ranked by descending information gain, and the top 5 were selected as the reduced feature set. All categorical features were one-hot encoded for modelling,

since neither logistic regression nor scikit-learn's decision tree can use string categories, and a numeric encoding would falsely imply an ordering between categories that don't have one.

Two classifiers logistic regression and a decision tree (with max depth=5 to prevent overfitting) were each trained twice, once on the full one-hot-encoded feature set (117 columns), and once only on the top-5 features by information gain (39 columns after one-hot encoding). All four resulting models were evaluated on the same held out test set using accuracy, precision, and recall (with 'poisonous' being the positive class, since a false negative would be the dangerous kind of error), and their confusion matrices were compared.

As a final cross check, the information-gain ranking (computed independently, before any model was trained) was compared against the trained decision tree's feature importances and its actual first split, to see whether the two independent measures of feature usefulness agree.

5. Results / Output

The entropy of the target on the training set was 0.9991 bits (out of a maximum of 1.0), confirming that the classes were nearly perfectly balanced.

Rank	Feature	Information Gain (bits)
1	odor	0.9028
2	spore-print-color	0.4849
3	gill-color	0.4150
4	ring-type	0.3169
5	stalk-surface-above-ring	0.2885
6	stalk-surface-below-ring	0.2726
7	stalk-color-above-ring	0.2539
8	stalk-color-below-ring	0.2441
9	gill-size	0.2243
10	population	0.2023

11	bruises	0.1879
12	habitat	0.1512
13	stalk-root	0.1334
14	gill-spacing	0.1020
15	cap-shape	0.0462

5.1 Information Gain Ranking

Rank	Feature	Information Gain (bits)
16	ring-number	0.0371
17	cap-color	0.0342
18	cap-surface	0.0264
19	veil-color	0.0240
20	gill-attachment	0.0147
21	stalk-shape	0.0084
22	veil-type	0.0000

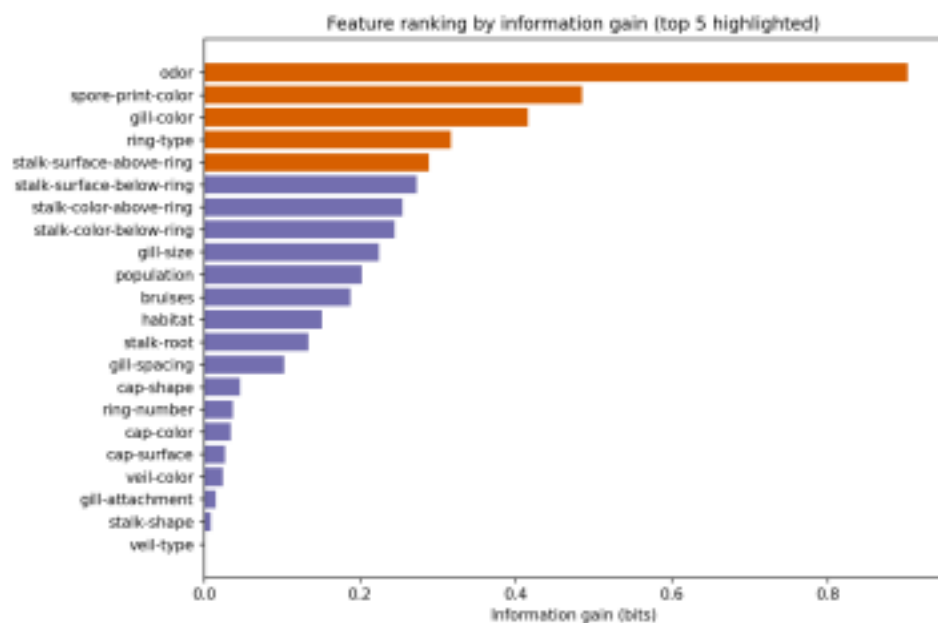


Figure 2. All 22 features ranked by information gain; the top 5 (highlighted) were used for the reduced models.

5.2 Model Performance

Model	Feature Set	Accuracy	Precision	Recall
Logistic Regression	Full (22)	0.9995	1.0000	0.9990
Logistic Regression	Top-5	0.9985	1.0000	0.9969
Decision Tree	Full (22)	0.9990	1.0000	0.9980
Decision Tree	Top-5	0.9985	1.0000	0.9969

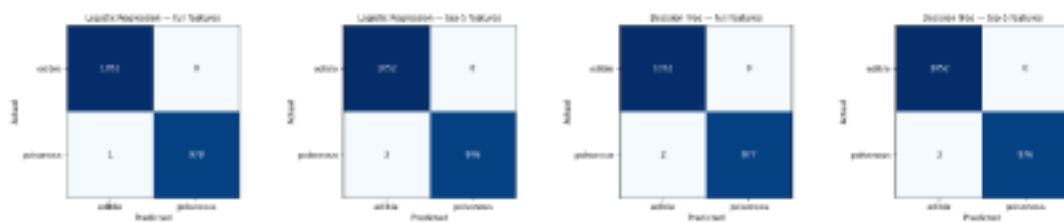


Figure 3. Confusion matrices for all four models (test set, $n = 2,031$).

5.3 Decision Tree Structure

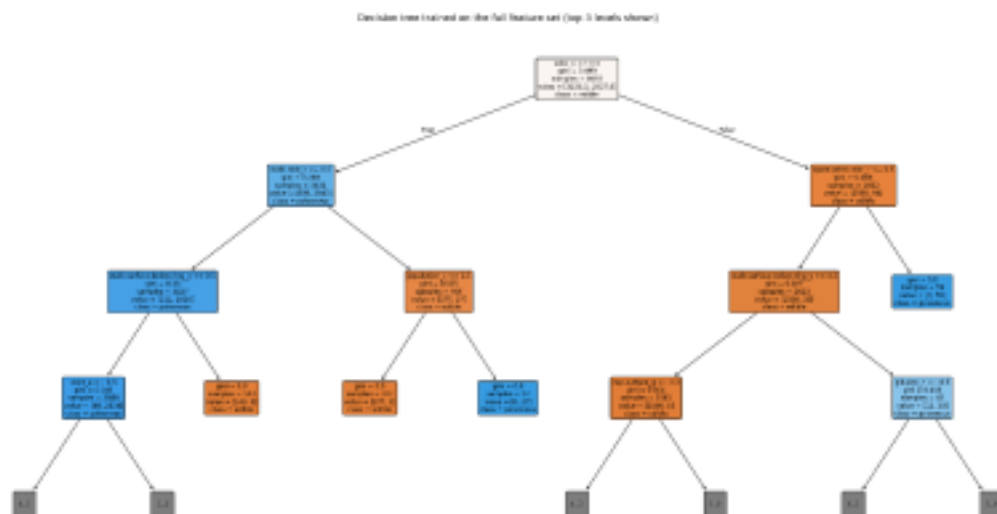


Figure 4. Top 3 levels of the decision tree trained on the full feature set. The root node

splits on 'odor'.

5.4 Logistic Regression Coefficients

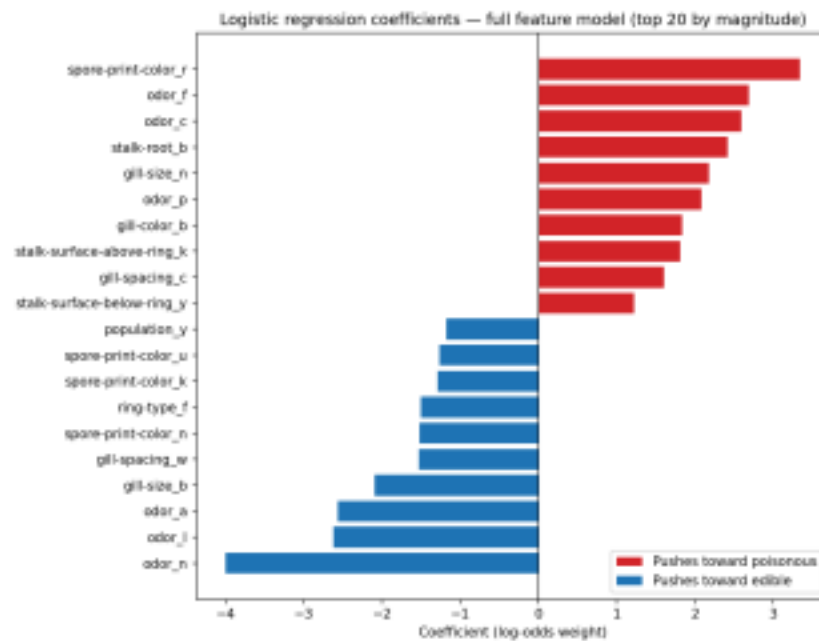


Figure 5. The 20 largest-magnitude coefficients from the full-feature logistic regression model.

5.5 Sigmoid Output on Real Test Data

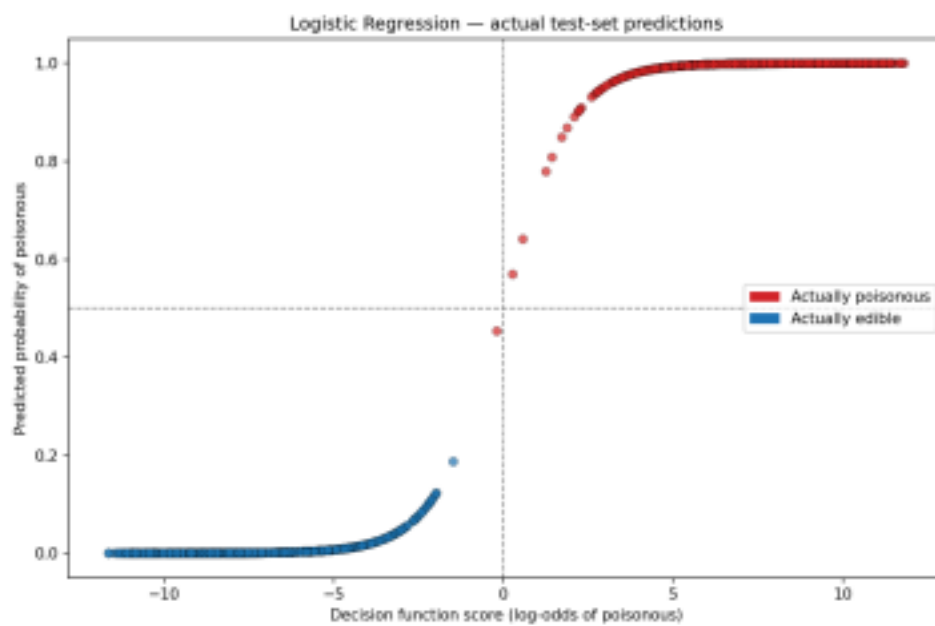


Figure 6. Each test-set mushroom's decision-function score (log-odds) plotted against its predicted probability of being poisonous, colored by a true label.

6. Analysis and Discussion

The most important finding is that reducing from 22 to the top 5 features (by information gain) cost little in accuracy: from 0.9995 to 0.9985 for the logistic regression (one or two additional wrong mushrooms out of 2,031 tested mushrooms), and 0.9990 to 0.9985 for the decision tree. This corroborates our hypothesis that information gain is an effective and computationally cheap way to select features for this dataset. The cross checking of the information gain ranking and the feature used to implement the decision tree's own strategy adds credibility to this hypothesis: the 'odor' feature was top ranked (information gain 0.9028 bits; almost twice that of the second ranked feature) and was in fact the feature chosen by the decision tree as its root split feature, simply by virtue of the tree's own greedy algorithm. Table 3 below compares the rankings from the two approaches for the top 10 features (the full 22 feature comparison is shown in Appendix A).

Feature	IG Rank	Information Gain	Tree Importance Rank	Tree Importance
odor	1	0.9028	1	0.6642
spore-print-color	2	0.4849	4	0.0369
gill-color	3	0.4150	15	0.0000
ring-type	4	0.3169	15	0.0000
stalk-surface-above-ring	5	0.2885	15	0.0000
stalk-surface-below-ring	6	0.2726	3	0.0992
stalk-color-above-ring	7	0.2539	15	0.0000
stalk-color-below-ring	8	0.2441	15	0.0000

Feature	IG Rank	Information Gain	Tree Importance Rank	Tree Importance
gill-size	9	0.2243	6	0.0056
population	10	0.2023	5	0.0166

The two rankings have an almost identical top (odor first in both), but diverge rapidly further down the list several features that have moderate information gain (e.g. gill-color, ring-type) are assigned zero importance by the trained tree. This is to be expected rather than contradictory once a tree has split on a strong feature like odor, many of the mushrooms are already perfectly separated, and so later features never get the opportunity to be used for a split, even if they possess real information gain on the entire dataset. Information gain is a measurement of a feature's usefulness in isolation, but a tree's feature importance is a measurement of usefulness given everything the tree has already split on before it, the two are related, but not identical.

It is also worth noting a methodological subtlety scikit-learn's `DecisionTreeClassifier` has a default criterion of 'Gini impurity', rather than entropy/information gain (the criterion which the ID3 algorithm and this project's background research describe). Gini impurity and entropy usually agree on which feature is most useful, though they are not strictly mathematically identical, hence why both point to 'odor' in this case, but the comparison in Table 3 shouldn't be read as a direct theoretical comparison between the two, so much as "two closely related but distinct criteria agree strongly". Passing `criterion='entropy'` when constructing the tree would make the comparison a closer theoretical match.

The logistic regression coefficients (Figure 5) give a third, independent perspective. The largest single coefficient pushing toward 'edible' was `odor = none`, followed by `odor = anise` and `odor = almond`; the largest coefficients pushing toward 'poisonous' were `spore_print_color = green` and several foul/creosote/pungent odor categories. This is the same feature (odor) and the same broad pattern (pleasant/no smell → edible, foul smell → poisonous) that both the information-gain ranking and the decision tree converged on independently — three different methods, separately, arrived on (roughly) the same answer.

Figure 6 shows why the accuracy numbers are all so high — the model's predicted probability is tightly bunched around 0 and 1, with almost no test mushrooms near the ambiguous 0.5 decision boundary. This means that the two classes are nearly linearly separable using only a handful of categorical cues, which is precisely why both the full feature and reduced feature models perform so similarly — there simply isn't much accuracy left on the table for extra features to capture, once the informative ones have been added.

7. Limitations

- The dataset's classes are almost perfectly separable using a small number of categorical features, making this a comparatively easy test case; in a dataset with more overlapping classes or more weakly informative features (i.e. ones that don't add much information gain) would be a tougher stress test for the claim that an information-gain cutoff discards features that matter.
- The decision tree was using scikit-learn's default of Gini impurity rather than entropy, so the comparison in Section 6 is not a theoretically perfect comparison (though they agree in practice here).
- The top-k value ($k = 5$) was chosen manually, and not through a formal method like locating an elbow in the information-gain curve or cross validating over a few possible values of k .
- The results are from a single train/test split (`random_state = 42`), and a different split or k -fold cross validation could change the exact accuracy figures slightly, though the overall conclusion is unlikely to change given how large the accuracy margin already is.
- Missing stalk root values were treated as their own separate category, rather than using a more sophisticated missing values imputation.
- The conclusion is specific to this dataset and shouldn't be assumed to generalize to feature selection problems in general, or ones with numeric (non-categorical) features.

8. Conclusion

This project sought to test the claim that ranking features by information gain and discarding the rest yields a model with accuracy comparable to using every feature, on the mushroom edibility dataset. The answer is yes and the margin is small enough to call the trade off essentially free: the full feature logistic regression reached 99.95% accuracy, versus 99.85% for the top 5 feature version, and the full feature

decision tree reached 99.90% vs. 99.85% for its top 5 feature version. In absolute terms, dropping 17 of the 22 original features cost at most one or two additional misclassifications out of 2,031 test examples, across both model types.

The cross check built into the methodology only strengthened the point the information gain ranking (computed independently of any trained model) put odor first by wide margin (0.90 bits, against 0.48 for the second ranked feature, `spore_print_color`) and the decision tree (trained separately from being told this ranking), chose odor as its very first split.

The logistic regression coefficients told the same story from a third angle: the largest weights pushing toward 'poisonous' were `spore_print_color` = green and specific foul/creosote/pungent odor categories, and the largest weights pushing toward 'edible' were the no odor, almond, and anise categories. Three independent methods, an information theoretic ranking, a tree's greedy splits, and a linear model's learned weights converged on the same handful of features, which is a stronger form of evidence than any of the three individually could have provided.

That said, this is a favourable case for information gain, not necessarily a representative one, as discussed in the Limitations section. The mushroom edibility classes turn out to be almost perfectly separable using a small number of categorical features, which is precisely what allowed every model (full and reduced) to clear 99.8% accuracy, and is why the logistic regression's coefficients ended up as large as they did. A dataset where the signal is spread out over many weakly informative features (which would have low information gain), or where classes overlap more, would be a tougher test of the same idea, and the natural next step if this work is continued.

Practically, the result still matters: a 5 feature model that loses under 0.1% accuracy is far cheaper to compute, easier to explain to a non-technical user, and easier to audit than a 22-feature (117-column, one-hot-encoded) one. Relevant for something like a field guide app, where a forager needs a handful of visible, memorable cues (like smelly mushrooms or a particular spore print colour) rather than a full taxonomic checklist.

9. References

1. UCI Machine Learning Repository — Mushroom Data Set.
2. Quinlan, J. R., Induction of Decision Trees (ID3 algorithm).
3. scikit-learn documentation — LogisticRegression, DecisionTreeClassifier.

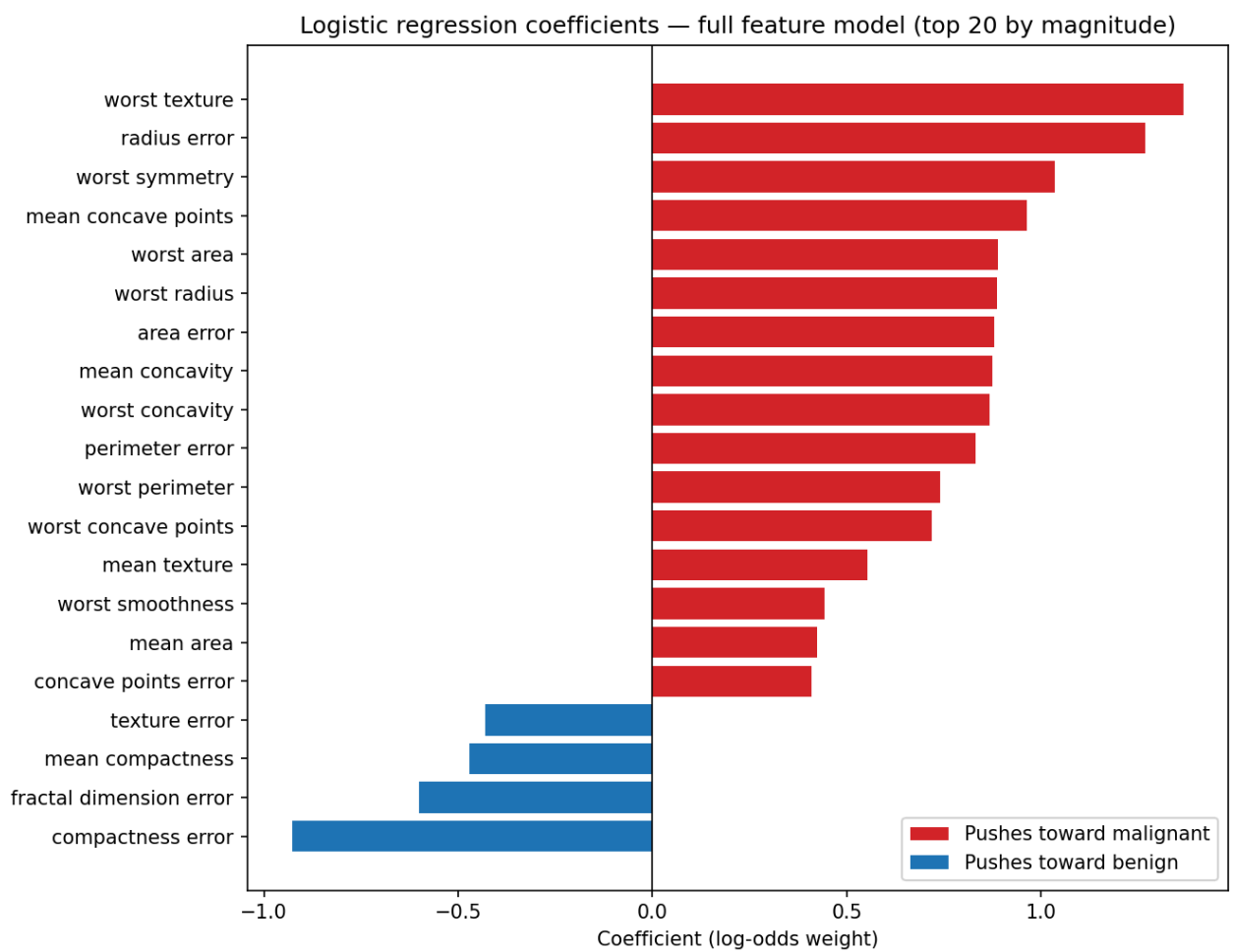
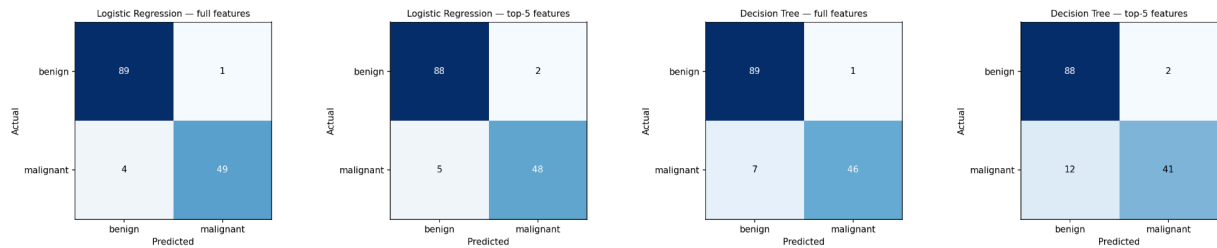
10. Appendices

Appendix A: Full Information-Gain vs. Tree-Importance Comparison

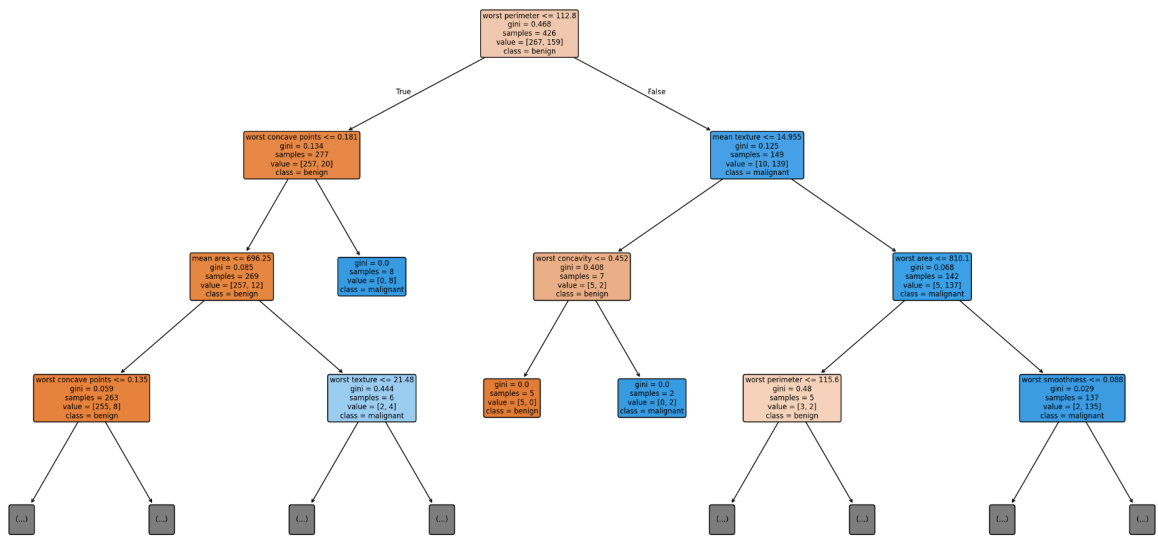
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gill-size	9	0.2243	6	0.0056
population	10	0.2023	5	0.0166
bruises	11	0.1879	15	0.0000
habitat	12	0.1512	15	0.0000
stalk-root	13	0.1334	2	0.1748
gill-spacing	14	0.1020	15	0.0000
cap-shape	15	0.0462	8	0.0007
ring-number	16	0.0371	15	0.0000
cap-color	17	0.0342	15	0.0000
cap-surface	18	0.0264	7	0.0020
veil-color	19	0.0240	15	0.0000
gill-attachment	20	0.0147	15	0.0000
stalk-shape	21	0.0084	15	0.0000
veil-type	22	0.0000	15	0.0000

[Code and video](#) - python files and video submission.

Graphs for breast cancer dataset(same program was run)



Decision tree trained on the full feature set (top 3 levels shown)



Model	Accuracy	Precision	Recall
Logistic Regression — full features	0.965034965034965	0.98	0.9245283018867925
Logistic Regression — top-5 features	0.951048951048951	0.96	0.9056603773584906
Decision Tree — full features	0.9440559440559441	0.9787234042553191	0.8679245283018868
Decision Tree — top-5 features	0.9020979020979021	0.9534883720930233	0.7735849056603774

